FAULT CURRENT STUDIO - ENGINEERING REPORT
BROWSER EXPORT - ENGINEERING REVIEW REQUIRED

===== SUMMARY =====
Project: Fault Current Studio advanced study
Method: peakscreen
Model status: BOUNDED_METHOD_C_PEAK_SCREEN
Captured UTC: 2026-10-26T06:04:34.862Z
Warning: ENGINEERING SCREENING ONLY - independently verify actual manufacturer, utility, installation, equipment settings and standards edition before professional use.
Report data provenance: Entered study inputs, isolated calculated method outputs. No external OEM certification or auto-selected breaker/PPE.

===== STUDY INPUTS =====
Input path: nominalLLV | Entered value: 13800
Input path: voltageFactor | Entered value: 1.1
Input path: frequencyHz | Entered value: 60
Input path: faultBus | Entered value: F
Input path: buses[0] | Entered value: F
Input path: buses[1] | Entered value: A
Input path: buses[2] | Entered value: B
Input path: branches[0].id | Entered value: feeder-A
Input path: branches[0].from | Entered value: A
Input path: branches[0].to | Entered value: F
Input path: branches[0].rOhm | Entered value: 0.06
Input path: branches[0].xOhm | Entered value: 0.12
Input path: branches[1].id | Entered value: feeder-B
Input path: branches[1].from | Entered value: B
Input path: branches[1].to | Entered value: F
Input path: branches[1].rOhm | Entered value: 0.04
Input path: branches[1].xOhm | Entered value: 0.13
Input path: sources[0].id | Entered value: utility-1
Input path: sources[0].bus | Entered value: A
Input path: sources[0].rOhm | Entered value: 0.08
Input path: sources[0].xOhm | Entered value: 0.42
Input path: sources[0].angleDeg | Entered value: 0
Input path: sources[0].emScale | Entered value: 1
Input path: sources[1].id | Entered value: generator-2
Input path: sources[1].bus | Entered value: B
Input path: sources[1].rOhm | Entered value: 0.11
Input path: sources[1].xOhm | Entered value: 0.6
Input path: sources[1].angleDeg | Entered value: 0
Input path: sources[1].emScale | Entered value: 1

===== EQUIPMENT DUTY =====
Calculated output path: status | Value: BOUNDED_METHOD_C_PEAK_SCREEN
Calculated output path: scope | Value: Bounded IEC 60909 equivalent-frequency Method C screen: one voltage level, documented passive positive-sequence network, identical source EMFs, 3PH bolted fault. Not a full IEC 60909-0:2026 calculation or breaker-making-duty certification.
Calculated output path: faultType | Value: 3PH_BOLTED
Calculated output path: faultBus | Value: F
Calculated output path: nominalLLV | Value: 13800
Calculated output path: voltageFactor | Value: 1.1
Calculated output path: frequencyHz | Value: 60
Calculated output path: equivalentFrequencyHz | Value: 24
Calculated output path: drivingPointNominalOhm.rOhm | Value: 0.07332704773129058
Calculated output path: drivingPointNominalOhm.xOhm | Value: 0.3106022392457279
Calculated output path: drivingPointNominalOhm.magnitudeOhm | Value: 0.31914041886518757

Calculated output path: drivingPointEquivalentFrequencyOhm.rOhm | Value: 0.07313849499070618
Calculated output path: drivingPointEquivalentFrequencyOhm.xOhm | Value: 0.12457118808524564
Calculated output path: drivingPointEquivalentFrequencyOhm.magnitudeOhm | Value: 0.14445490767182403
Calculated output path: nominalRX | Value: 0.23608022887845018
Calculated output path: methodCRX | Value: 0.23484883178815502
Calculated output path: kappaMethodC | Value: 1.5044460514914157
Calculated output path: initialSymmetricalA | Value: 27461.821092616654
Calculated output path: initialSymmetricalKA | Value: 27.461821092616653
Calculated output path: peakMakingScreenA | Value: 58427.990522482636
Calculated output path: peakMakingScreenKA | Value: 58.42799052248264
Calculated output path: nodalRelativeResidual | Value: 8.950904182623615e-16
Calculated output path: limitations[0] | Value: Positive-sequence passive equivalent impedances, one nominal voltage level and coincident source EMFs only
Calculated output path: limitations[1] | Value: IEC equipment impedance correction factors and automatic c selection are NOT implemented
Calculated output path: limitations[2] | Value: Reactances assumed linearly proportional to frequency and resistances constant between f and 0.4f
Calculated output path: limitations[3] | Value: No transformer vector groups/taps, generator decrement, converter or motor dynamic contributions
Calculated output path: limitations[4] | Value: Not a validated IEC 60909 breaker-making or interrupting duty calculation; no label or device approval

===== ASSUMPTIONS & QA =====
Evidence / limitation: Release status | Value: BOUNDED METHOD, NOT A STANDARDS/DEVICE APPROVAL
Evidence / limitation: Warning | Value: ENGINEERING SCREENING ONLY - independently verify actual manufacturer, utility, installation, equipment settings and standards edition before professional use.
Evidence / limitation: Study method | Value: peakscreen
Evidence / limitation: Timestamp | Value: 2026-10-26T06:04:34.862Z
Evidence / limitation: outputs.scope | Value: Bounded IEC 60909 equivalent-frequency Method C screen: one voltage level, documented passive positive-sequence network, identical source EMFs, 3PH bolted fault. Not a full IEC 60909-0:2026 calculation or breaker-making-duty certification.
Evidence / limitation: outputs.limitations[0] | Value: Positive-sequence passive equivalent impedances, one nominal voltage level and coincident source EMFs only
Evidence / limitation: outputs.limitations[1] | Value: IEC equipment impedance correction factors and automatic c selection are NOT implemented
Evidence / limitation: outputs.limitations[2] | Value: Reactances assumed linearly proportional to frequency and resistances constant between f and 0.4f
Evidence / limitation: outputs.limitations[3] | Value: No transformer vector groups/taps, generator decrement, converter or motor dynamic contributions
Evidence / limitation: outputs.limitations[4] | Value: Not a validated IEC 60909 breaker-making or interrupting duty calculation; no label or device approval